Q1) i. Conditional Branches (BEQ):

- Stage-1 (IF/ID/RR): The instruction is fetched from instruction memory. It is decoded to identify it as BEQ instruction. 'rs1' and 'rs2' are read from register file. The 'imm' value is extracted and sign-extended.

- Stage2 (EX/MEM/WB): ALU compares rs1 & rs2. If ALU zero flag is set, the branch target address is calculated by finding 'PC + offset'. If not equal, then PC ← PC+4.

ii. Unconditional branch (JAL):

- Stage-1: Instruction is fetched and decoded as 'jal'. The imm value (jump offset) is sign-extended.

- Stage-2: The return address (PC+4) is written back to 'rd'. PC ← PC + offset.

iii. ALU (ADD):

- Stage1: R-type add instruction. 'rs1' and 'rs2' are read from RF.

- Stage2: ALU performs 'add' on rs1 & rs2. Result is stored in 'rd'.

iv. Load (LW):

- Stage-1: Instruction is fetched and decoded as LW. The base address register 'rs1' is read from RF and immediate offset is sign-extended.

- Stage-2:- ALU adds 'rs1' value & sign-extended imm value to calculate mem address. The DMEM is accessed and the 32-bit word is written back to rd.

v. Store (SW):-

- Stage1:- Both base address 'rs1' and register containing the data to be stored 'rs2' are read from RF. The 'imm' offset is sign-extended.
- Stage 2:- ALU calculates effective mem address by adding 'rs1' and 'imm' offset. Value from 'rs2' is written to DMEM at calculated address.